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GLOBAL SYSTEM FOR
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Digital cellular telecommunications system (Phase 2); Types of Mobile Stations (MS) (GSM 02.06 version 4.5.2)

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Foreword

This European Telecommunication Standard (ETS) has been produced by the Special Mobile Group (SMG) of the European Telecommunications Standards Institute (ETSI).

This ETS defines typical attributes of different types of Mobile Stations (MS) used with GSM and DCS 1 800 PLMNs within the European digital cellular telecommunications system (Phase 2).

The specification from which this ETS has been derived was originally based on CEPT documentation, hence the presentation of this ETS may not be entirely in accordance with the ETSI/PNE Rules.

Transposition dates	
Date of adoption of this ETS:	5 December 1997
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1 Scope

The purpose of this ETS is to describe typical attributes of different types of Mobile Stations (MS) to be used on GSM PLMNs. The term GSM PLMN refers to a GSM, DCS 1 800 or any PLMN based on the GSM specification, irrespective of the frequency band used.

Manufacturers and customers may choose any appropriate combination of these attributes in order to fulfil their need while utilizing the services offered through a GSM or a DCS 1 800 PLMN. This is not an exhaustive list of attributes or types of Mobile Station. Type approval of Mobile Stations is addressed in GSM 11.10.

1.1 Normative references

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] GSM 01.04 (ETR 100): "Digital cellular telecommunications system (Phase 2); Abbreviations and acronyms".

[2] GSM 04.03 (ETS 300 552): "Digital cellular telecommunications system (Phase 2); Mobile Station - Base Station System (MS-BSS) interface Channel structures and access capabilities".

[3] GSM 04.08 (ETS 300 557): "Digital cellular telecommunications system (Phase 2); Mobile radio interface layer 3 specification".

[4] GSM 05.05 (ETS 300 577): "Digital cellular telecommunications system (Phase 2); Radio transmission and reception".

[5] GSM 11.10 (ETS 300 607): "Digital cellular telecommunications system (Phase 2); Mobile Station (MS) conformity specification".

1.2 Abbreviations

Abbreviations used in this document are listed in GSM 01.04.

2 General

The term Mobile Station is taken to mean equipment necessary to access GSM PLMN telecommunication services. It includes the mobile termination and also may include Terminal Equipments and Terminal Adaptors. Some of the characteristics of the configuration in use at any time will be reflected in the Mobile Station Class Mark (see GSM 04.08). A Mobile Station is characterized by a set of attributes defined in clause 3.

3 Mobile Station attributes

The list of Mobile Station attributes is as follows:

- MS service access configuration;
- MS access capability;
- Modes of Use.

3.1 MS service access configuration

Service access configuration must suit the requirements of the customer and will depend on the combination of tele-, bearer and supplementary services to be used. The actual configuration will depend on the manufactuers' implementation and may comprise a single unit or a mobile termination unit with additional Terminal Equipment and/or Terminal Adaptors.

3.2 MS access capability

The Mobile Station access capability is defined in GSM 04.03 Channel Structures and Access Capability, and describes the limitation put on the simultaneous provision at a given time of tele- or bearer services.

3.2.1 Frequency Bands

Three frequency bands are defined:

i) Standard GSM Band;

ii) Extended GSM Band (includes standard band);

iii) DCS 1 800 Band.

Mobile Stations may support one or more of these bands.

A Mobile Station which supports more than one band and the functionality below is defined as a Multi-band Mobile Station.

> The Multi-band mobile station has functionality to perform handover, channel assignment, cell selection and cell re-selection between all its bands of operation within a PLMN, i.e. when one PLMN code is used in all bands. In addition it has the functionality to make PLMN selection, in manual or automatic mode, in all its bands of operation. The multi-band mobile station shall meet all requirements specified for each individual band. In addition it shall meet the extra functional requirements for multi band mobile stations.

An MS capable of GSM and DCS 1 800 operation is also allowed where the band of operation is selected by means of an internal manual or automatic procedure. This type of MS cannot use the GSM and DCS 1 800 modes simultaneously and does not support handover, channel assignment, cell selection or reselection between the bands of operation. Once the frequency band has been selected, the MS shall comply with all the specifications relevant to that band. The network does not provide any more functionality to support this type of MS than that required for single band MS.

3.2.2 Full Rate/Half Rate Services

Both Full Rate and Half Rate services are specified. For each basic service, they may be supported by a mobile station on full rate, full and half rate, or not at all. MSs supporting services using signalling modes only are also permitted.

> NOTE: The network does not provide any functionality to manage dual mode GSM and DCS-1 800 MS.

3.2.3 Speech codecs

Speech teleservices may be provided using the Full Rate (full rate, version 1), Enhanced Full Rate (full rate, version 2) or Half Rate (half rate, version 1) speech codecs. For speech services, the allowed MS configurations are:

- Full Rate (full rate, version 1) speech codec only.
- Full Rate (full rate, version 1) plus any combination of the Enhanced Full Rate (full rate, version 2) and Half Rate (half-rate, version 1) speech codecs.

3.3 Definition of a hand-held MS

In this standard, a handheld station is an MS where the used antenna is directly attached to the portion of the equipment containing the acoustic transducers for speech (this condition applies only to speech mobile stations).

3.4 Definition of a vehicular mounted MS

In this standard, a vehicular mounted MS is an MS where the used antenna is physically mounted to the outside of a vehicle. Vehicles include, for example, trucks, buses, trains and ships.

4 Mobile station output and power control

GSM and DCS MS are categorized into classes according to their maximum output power as defined in GSM 05.05. The mean output power for each of these classes is as follows:

MS class	Nominal maximum mean power (milliwatts)	
	Full rate	Half rate
GSM class 2	960	500
GSM class 3	600	312
GSM class 4	240	125
GSM class 5	96	50
DCS class 1	120	62
DCS class 2	30	16
DCS class 3	480	250

A multi-band MS has a combination of power classes, one from each of the bands of operation in the table above. Any combination can be used.

GSM and DCS 1 800 Mobile stations must be capable of reducing transmitter output power on command from a base station. See GSM 05.05.

5 Configuration possibilities

Any given configuration will be a combination of attributes mentioned in clause 3 and may include additional features e.g. payphones.

A GSM handheld MS shall only be of GSM class 4 or 5. A DCS handheld MS shall only be DCS of class 1 or 2. A handheld multi-band MS must meet the above requirements on each band supported by the MS.

A DCS MS of DCS class 3 shall only be vehicular mounted.

Annex A (informative): Change history

Change history					
SMG No.	**TDoc. No.**	**CR. No.**	**Suclauses affected**	**New version**	**Subject/Comments**
SMG#22	311/97	A003	3.2.2 3.2.3	4.5.0	Allowed Speech Codec Configuration in the MS

History

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